INTERNAL · FACT-CHECK OF THE PITCH DECK AND BUSINESS PLAN · 11 NOV 2026

loop. numbers check

We checked every claim and number in the deck, the script and the business plan against real sources: official statistics, company filings and industry benchmarks. The arithmetic in our plan is correct. The market numbers are mostly right. But several of our student-side assumptions are best-case or beyond, and the revenue timing is off. The university licence is where the unit economics actually work.

market
mostly correct; 2 fixes

student LTV:CAC
3.5:1 → realistic ≈ 0.3:1

licence LTV:CAC
≈ 3.5:1, holds up

year-3 revenue
€1.2M → realistic ≈ €0.4M

1 · market claims

IN THE DECK	VERDICT	REAL FIGURE AND SOURCE
EU: 19.1M tertiary students (2024)	**correct**	19.1M in 2024, newest year available (Eurostat). Note: "tertiary" includes vocational higher education, so it is broader than "university students". Keep the word "tertiary".
Spain: 1.83M university students (2024–25)	**close, wrong source**	The latest official totals are **1.76M** (2023–24, Ministry of Science, Innovation and Universities) and **1.875M** (2025–26, Ministry). Replace Cadena SER/Wikipedia with the Ministry figure: **"1.88M (2025–26, Ministry)"**.
IE University: 10,000+	**correct**	"Over 10,000 Bachelor's and Master's students from 160 countries" (IE University news, Sept 2025). Cite IE, not Wikipedia.
Market ≈ €478M (19.1M × €25)	**method too generous**	This assumes every student pays. Better: a ceiling of **€191M a year** if every EU tertiary student were covered by a €10 licence (Spain ≈ **€19M**). Third-party reference: global education-apps market USD 7.3B in 2025 (Straits Research, commercial estimate).
Programme licence = 1,500–2,000 students	**too big**	Spain has ≈ 1.38M undergraduate students across 3,300–4,500 degree programmes, so ≈ **300–400 students per programme**. 1,500 students is a whole **faculty or school**. Sell per faculty.
10 Spanish universities in year 1 (plan)	**room to grow**	Spain has 92 universities (2024–25): 50 public, 42 private (Ministry). The private ones are the faster first targets.

2 · student unit economics (B2C)

ASSUMPTION	OURS	REAL-LIFE	EVIDENCE
Price €4.99/mo · €29/yr	ok	ok, even cheap	Education-app median is $9.99/mo and $44.99/yr (RevenueCat 2026). Comparable apps: Knowunity Pro ≈ $4.49/mo, Brainly ≈ $24/yr, Vaia €12.49/mo.
Revenue per payer	€25/yr	**€20–24**	Prices include 21% Spanish VAT (€29 → €23.97 net). RevenueCat's median first-year revenue per education payer is $22.82.
Free → paid	6%	**2–4%**	The freemium median is 2.1% and the top quartile >4.5% (RevenueCat 2026); Western Europe is 2.0%. Duolingo, best in class, has 8.6% of monthly users paying after 14 years.
Cost per sign-up	€0.80	**€1.50–3.00**	€0.80 is realistic only for the first organic users. Paid installs in Europe cost ≈ $2.25 (Adjust). Ambassador programmes typically pay $5–10 per sign-up.
Gross margin	85%	**65–80%**	Duolingo's gross margin is 72.9% and falling because of AI costs. AI app companies run at ≈ 50–60%. A web app avoids app-store fees; Stripe is 1.5% + €0.25.

Paying lifetime	2.2 years	**≈ 1 year**	Only ≈ 8% of monthly subscribers are still paying after 12 months. Annual plans renew at 23–40%. AI apps churn ≈ 30% faster (RevenueCat 2026).

	LTV	CAC	LTV : CAC	PAYBACK
Our pitch (€25 × 85% × 2.2 yrs; €0.80 ÷ 6%)	€47	€13	3.5 : 1	8 mo
Best case (€24 × 80% × 1.5 yrs; €0.80 ÷ 5%)	€29	€16	1.8 : 1	10 mo
Realistic (€22 × 75% × 1 yr; €1.50 ÷ 3%)	**€16.50**	**€50**	**0.3 : 1**	**36 mo**

What this means: students alone do not pay back what it costs to acquire them. That is common in consumer edtech and fine for our model, as long as we say it: the free student app is our **acquisition channel and proof**, and university licences are the business. The student LTV:CAC of 3.5:1 should come off the slide.

3 · university licences (B2B)

ASSUMPTION	OURS	REAL-LIFE	EVIDENCE
€10 per student per year	€10	ok	Top Hat ≈ $10 per student per semester; Grammarly for Education ≈ $36 per user per year; Packback $39–49 per course. EU tools (Wooclap, Mentimeter, Kahoot) charge per teacher, not per student.
Licence size	€15–20k	**< €15k** net of VAT	Spanish public-sector rule: a "contrato menor" (fast, no tender) must be **below €15,000 excluding VAT**, for one year at most (Ley 9/2017, art. 118). Above that requires a public tender, which takes months. Price a faculty at ≈ €14.5k (1,450 students × €10) and start with private universities, which aren't bound by this law.
Sales cycle	pilot → licence	**6–18 months**	Higher-ed sales take 6–18 months; vendor registration alone takes 3–6 (EDUCAUSE 2026; ASU at SXSW EDU 2026).
CAC per licence	€6k	**€8–12k**	A salesperson at ≈ €59k fully loaded closing 5–8 deals a year gives €8–12k per deal.
Licences, year 1	6	**2–4**	3 pilots, then one term of piloting, then procurement: first paid licences arrive around months 9–18.
LTV : CAC	**6 : 1**	**3.5 : 1**	**€14.5k × 80% margin × 3 years = €34.8k LTV ÷ €10k CAC. Payback ≈ 10 months. A healthy B2B ratio (the floor is 3:1).**

4 · costs and funding

ASSUMPTION	OURS	REAL-LIFE	EVIDENCE
Cost per employee	€47.5k	**€55–60k**	Madrid software engineers average ≈ €38k gross; business developers €45–65k. Employer social security in 2026 is ≈ 32% on top.
AI & infrastructure	€15k / yr 1	ok, generous	Claude Opus 5.5 lists at $4 / $20 per million tokens (in/out): ≈ $0.16 to turn one lecture into a feed, ≈ $50–100 per programme per year. Students' interactive AI use and hosting are the real cost drivers.
€400k pre-seed for 18 months	€400k	ok	In range for European pre-seed rounds ($250k–1M). Add non-dilutive funding: **ENISA Jóvenes Emprendedores** (€25–75k loan) and **CDTI Neotec** (up to €250k grant; next call likely spring 2027).
Seed ≈ €1.5M at €25k MRR	€1.5M	ok, but tighter	Spanish seed rounds average ≈ €2.1M (Fundación Bankinter 2025), but European seed deals fell 44% year on year in Q1 2026, so investors want more traction.

5 · the model, corrected

Two fixes compound here. The realistic inputs above lower the totals. And our plan counted year-end students and licences as if they had paid all year: the same plan with mid-year timing makes €58k in year 1, not €115k.

REALISTIC BASE CASE	YEAR 1	YEAR 2	YEAR 3	PITCH NOW (YR 3)
Registered students (end of year)	10,000	40,000	120,000	200,000
Paying students (3% → 4%)	300	1,400	4,800	13,000
Faculty licences (end of year)	3	12	30	42
Student revenue (€22 net)	€3k	€19k	€71k	€351k
Licence revenue (€14.5k → €16k)	€22k	€113k	€336k	€840k
Revenue	**€25k**	**€131k**	**€407k**	**€1,191k**

Team (3 → 6 → 9 people × €57.5k)	€173k	€345k	€518k	€700k
AI & infra · marketing (€1.50/sign-up) · sales & ops	€55k	€140k	€280k	€450k
Result (EBITDA)	−€202k	−€354k	−€390k	+€41k
Monthly revenue at year end	€4k	€18k	€49k	—

- **Break-even** needs ≈ 58 faculty licences on a year-3 cost base: realistically **year 4–5**, not year 3.
- **€400k still covers 18 months:** year-1 loss + half of year 2 ≈ €379k, plus a ≈10% buffer and ENISA/Neotec on top.
- **Month-18 milestone** becomes ≈ **€11k monthly revenue and ≈ 8 paying faculties**, not €25k.
- Licences become ≈ 82% of revenue by year 3. The story is a **B2B company with a free student app**, and that's a stronger, more honest pitch.

6 · what to change, slide by slide

SLIDE	CHANGE
Business model	Keep the three tiers. Universities: "**€10 per student · faculty licence**". Replace "CAC €13 · LTV €47 · 3.5:1" with "**licence LTV : CAC 3.5 : 1** (€14.5k × 3 yrs × 80% ÷ €10k)" and "students: free, €1.50 per sign-up". Assumptions line: "3% upgrade · €22 net per payer · 1-year paying life · 75% margin".
Market	Spain "**1.88M (2025–26, Ministry)**"; IE "10,000+ (IE, 2025)"; EU "19.1M tertiary (Eurostat 2024)". Say "up to €191M a year in Europe at €10 per student", not €478M.
Financials	"**€25k → €131k → €407k**", licences ≈ 80% of revenue; break-even year 4–5 at ≈ 60 faculty licences. Headline e.g. "€400k revenue by year 3."
The ask	Keep €400k / 18 months. Months 6–12: "10,000 students, **3 paid faculty licences**". Months 12–18: "**≈ €11k monthly revenue, 8 faculties**, seed-ready". Add "+ ENISA / Neotec".
Go-to-market	Add: "**private universities first**, faculty licences under €15k".
Traction, problem, solution	Our own numbers (15 sign-ups, 7 testers, 3/5) are fine. Still missing: the real problem story and the top tester feedback.

One line for Q&A: "Students rarely pay enough to cover their own acquisition, so we don't rely on them to. The free app builds usage; faculties pay €10 per student, which is a 3.5 to 1 return on what it costs us to sign one."

sources

- Eurostat, Tertiary education statistics (2024 data): ec.europa.eu/eurostat/statistics-explained/index.php?title=Tertiary_education_statistics
- Ministry of Science, Innovation and Universities: university report, 22 Jan 2025 (1.76M, 2023–24) ciencia.gob.es/Noticias/2025/enero/informe-universidades-espanolas.html · CEUNE, 30 Jan 2026 (1.875M, 2025–26) ciencia.gob.es/Noticias/2026/enero/pleno-Consejo-CEUNE.html · universities and degrees 2024–25 ciencia.gob.es/Noticias/2025/junio/estadistica-universidades-centros-titulaciones-2024–2025.html
- IE University news, 8 Sep 2025: ie.edu/university/news-events/news/ie-university-opens-academic-year-4–new-bachelors-masters-programs-technology-design-political-science/
- RevenueCat, State of Subscription Apps 2026 (and Education report): revenuecat.com/state-of-subscription-apps-2026/ · revenuecat.com/state-of-subscription-apps-2026–education · TechCrunch, 10 Mar 2026, AI apps retention
- Duolingo Q3 2026 shareholder letter (SEC): sec.gov/Archives/edgar/data/0001562088/000162828026069474/
- Adjust benchmarks via Promodo (Europe cost per install) · Makai, student ambassador compensation · Mayfield and Bain Capital Ventures on AI gross margins · Stripe EEA pricing
- Top Hat (University of Iowa ITS) · Perusall pricing · Wooclap help centre · Grammarly for Education (CU Anschutz)
- Ley 9/2017 de Contratos del Sector Público, art. 118 (contrato menor < €15,000): congreso.es/es/cem/contratmenor
- EDUCAUSE 2026, higher-ed purchasing guide · Indeed / Payscale Madrid salaries · 2026 Spanish social-security rates (Asepeyo) · Anthropic API pricing (platform.claude.com) · Fundación Bankinter Startup Observatory 2025 · ENISA · CDTI Neotec 2026 (ciencia.gob.es)
- Straits Research, Education Apps Market (commercial estimate)

Some benchmarks come from industry blogs and aggregators, so treat them as directional. Our own pilot data (conversion, retention, cost per sign-up) should replace them as soon as we have it. Have the Communications team review the deck before it is shared outside the course.